

CANDIDATE
NAME

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INDEX NUMBER

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BIOLOGY

9648/03

Applications Paper and Planning Question
Paper 3

20 September 2013

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

All working for numerical answers must be shown.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Calculators may be used

For Examiner's Use	
1	
2	
3	
4	
5	
Total	72

Answer **all** the questions.

- 1** In a maternity ward at a local hospital, a mix-up involving three couples and three babies caused a lot of confusion. Based on phenotypic characteristics, the nurses were unable to correctly identify the parents of the babies. In order to solve the case, a scientist was called in to carry out a DNA test to identify the parents of the babies. The test was based on the principle that different individuals have a different number of repeating units at a particular locus in a chromosome.

(a) State the name of the test that was carried out.

[1]

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Chromosome 13 was isolated from the DNA samples that were obtained from the three couples and three babies and used for further analysis. The sequence below shows a segment of chromosome 13, which was used in the analysis where (TTAGGAT) is the repeating unit and n is the number of repeats.

5' ...GCTAAGTATTGCTCAAGA... (TTAGGAT) $_n$...GATAAATAACTGGCTAGTA...-3'
3' ...CGATTCATAACGAGTTCT... (AATCCTA) $_n$... CTATTTATTGACCGATCAT...-5'

The gel diagram in Fig. 1.1 shows the results of the DNA test obtained from each individual, Dad and Mom #1, #2 and #3, and baby A, B, and C.

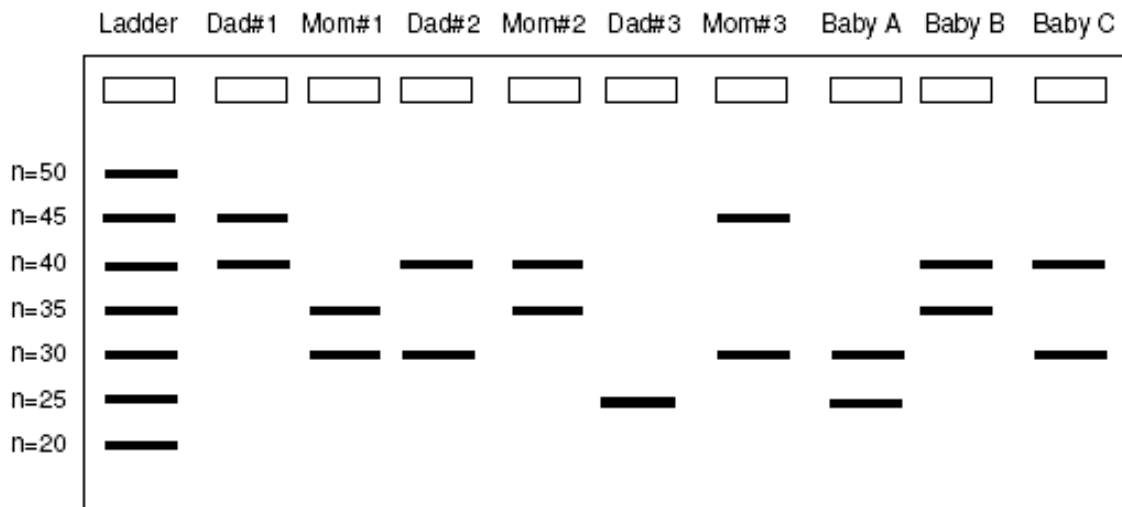


Fig. 1.1

(b) (i) Explain the purpose of the DNA ladder in Fig. 1.1.

[1]

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(ii) DNA is colourless.
State how the DNA bands in the gel can be made visible.

[1]

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(c) With reference to Fig. 1.1, explain why

(i) some bands in a lane are higher than others;

[2]

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(ii) the lane for Dad#3 has only one band;

[2]

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(iii) some lanes have two bands.

[2]

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(d) **(i)** With reference to Fig. 1.1, state and explain which couple is the parents of Baby A.

[2]

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(ii) Explain why results of paternity testing are not absolutely certain.

[2]

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[Total: 13]

- 2 One approach of gene therapy to treat cystic fibrosis uses viruses to deliver normal alleles of the CFTR gene into epithelial cells of the airways.

A team of researchers in the USA developed a new strain of non-pathogenic adeno-associated virus (AAV), AAV2.5T. Genes for the CFTR protein and the enzyme luciferase were inserted into the DNA of the viruses. Luciferase catalyses the production of a green fluorescent protein when luciferin is added.

The normal AAV strain and the AAV2.5T strain were added to cultures of epithelial cells from the airways. After adding luciferin, the number of cells that had taken up the viral genes was estimated using the intensity of the green fluorescence which developed. The results are shown in Fig. 2.1.

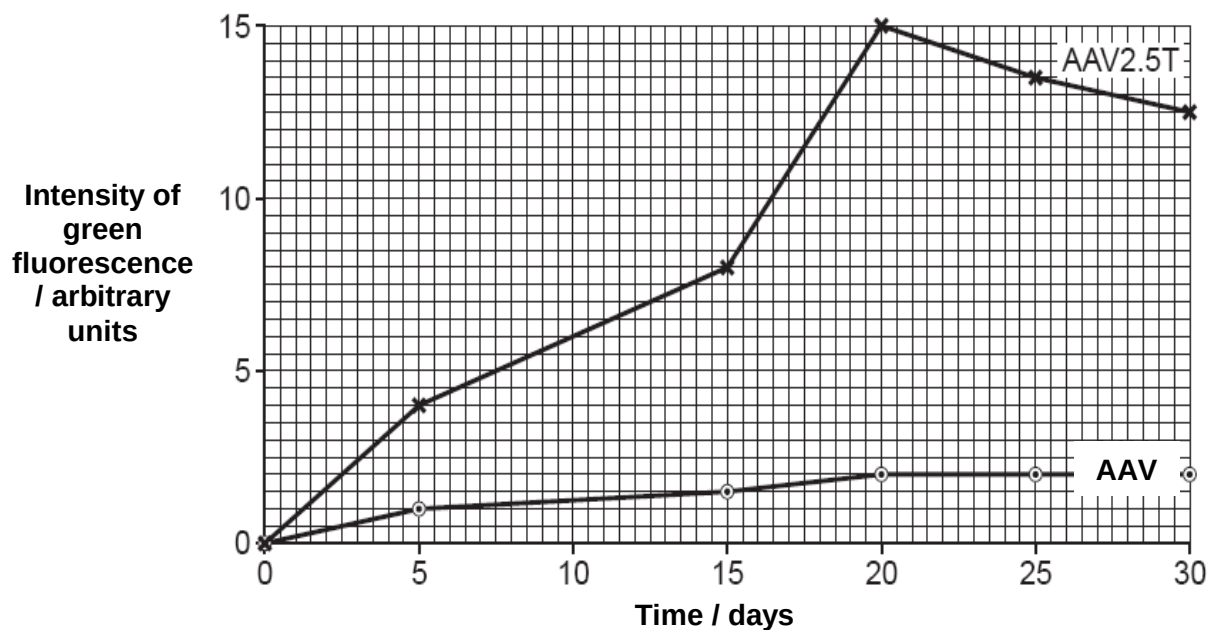


Fig. 2.1

- (a) Explain why the researchers inserted a gene for luciferase into the viral DNA.

[1]

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- (b) With reference to Fig. 2.1, compare the ability of the two viral strains, AAV and AAV2.5T, to infect epithelial cells of the airways.

[2]

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- (c) Suggest why a decrease in intensity of green fluorescence was detected in cells infected with AAV2.5T during the last 10 days. [2]

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- (d) Describe how delivering normal alleles of the CFTR gene into epithelial cells in the airways could relieve the symptoms of cystic fibrosis. [4]

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- (e) Researchers have started to use cDNA of the CFTR gene for more effective therapeutic results. Explain why cDNA of the normal CFTR gene is used instead of the normal CFTR gene. [2]

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- (f) AAV does not cause human disease, but because we are routinely exposed to this virus, 30 to 60 percent of people develop antibodies that neutralize AAV if it enters the circulation.

To extend the potential benefits of gene therapy to a broader population, researchers have produced a bioengineered decoy, which are empty AAV capsids disabled of their ability to enter target cells.

Suggest how using both the decoy and vector at the same time would improve the success rate of the gene therapy.

[3]

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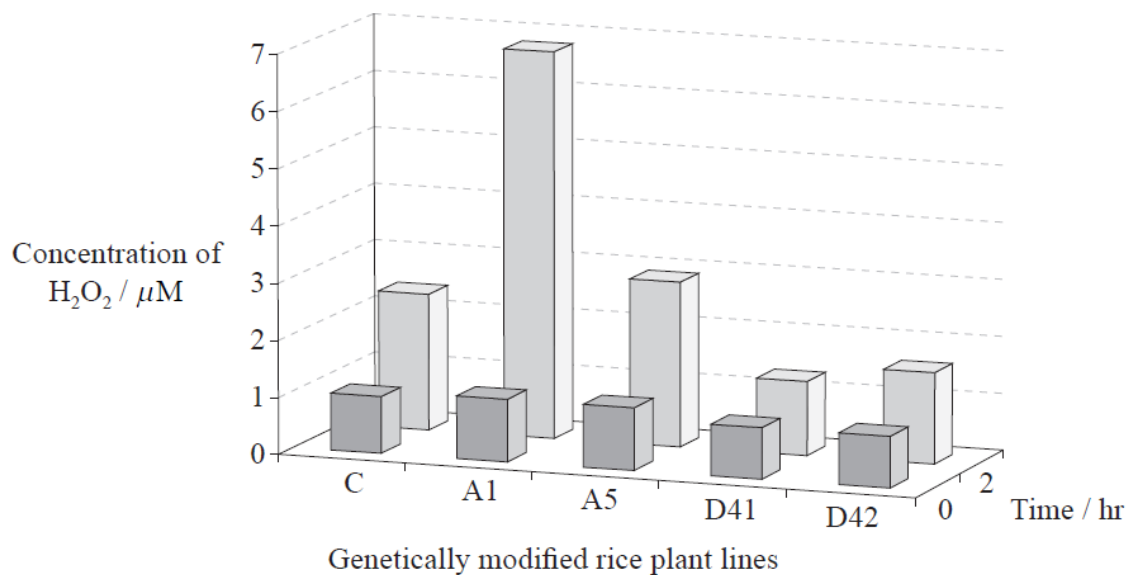
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[Total: 14]

- 3 Plants have developed defense mechanisms against pathogens such as bacteria, fungi, and viruses. Chemicals released by these pathogens can trigger a defense response in infected plant cells. For example, the production of hydrogen peroxide (H_2O_2) which reacts with pathogen membranes and cellular chemicals, eventually kills both the cell and the pathogen.

The *OSRac1* gene was isolated and introduced into a number of rice plant (*Oryza spp.*) lines to study its role in disease resistance of plants to Blast fungus. Experiments were carried out to see if the *OSRac1* gene was part of the signalling pathway for hydrogen peroxide production.

A control and four other genetically modified rice plant lines were exposed to chemicals known to initiate a defense response and the production of hydrogen peroxide. The results are shown in Fig. 3.1 below.



Key: C: control

A1 and A5: rice plants with the *OSRac1* gene always turned on

D41 and D42: rice plants with the *OSRac1* gene suppressed

Fig. 3.1

- (a) With reference to Fig. 3.1, compare the changes in H_2O_2 production in the control and genetically modified plants after the chemical was applied and conclude if *OSRac1* gene is involved in disease resistance. [4]

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- (b) Suggest **two** possible concerns about the use of transgenic plants with the disease resistance gene. [2]

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- (c) Edible vaccine against cholera using rice was developed by a team of Japanese scientists by cloning a gene from cholera bacteria, *Vibrio cholerae* into the genome of the Kitaake rice plant. The gene expressed a subunit of the disease-causing cholera toxin B. The steps are shown in Fig. 3.2.

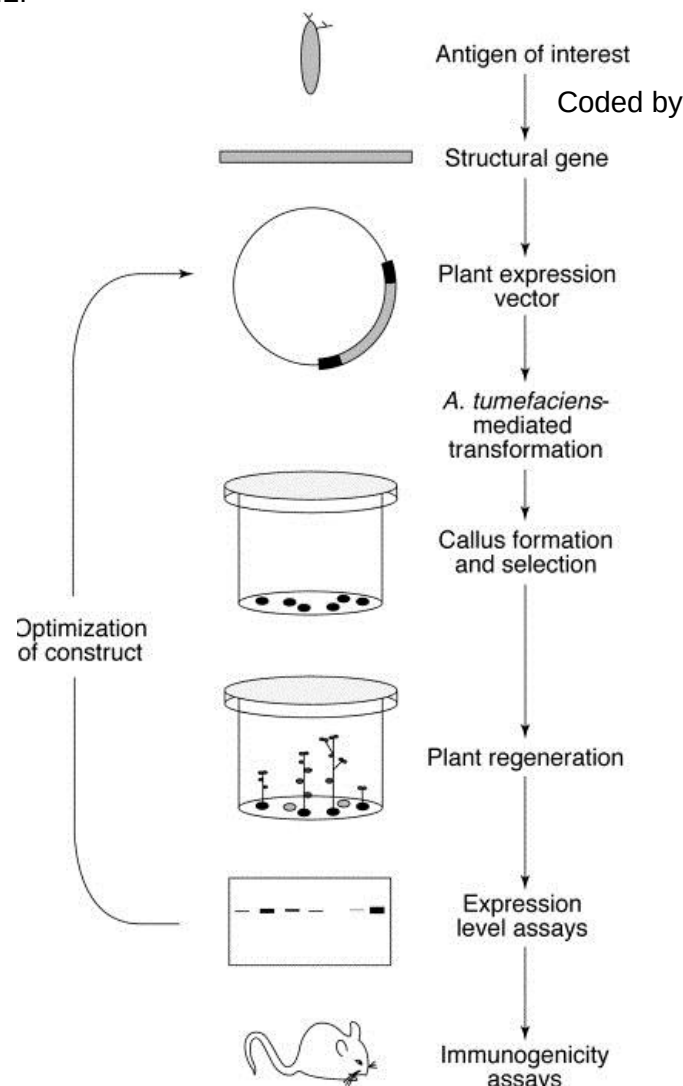


Fig. 3.2

- (i) Describe how the antigen gene inserted into rice DNA genome would result in successful vaccination once the rice is eaten. [2]

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- (ii) Explain why it is necessary to form a callus in order to grow genetically modified rice. [3]

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- (d) Edible vaccines have been made using other plants like tomato and banana. Suggest the advantages of using rice over other plants. [2]

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[Total: 13]

4 Planning question

DNA is the genetic material in all living organisms. DNA from an external source can be taken up by bacteria by a process known as transformation.

Plan an experiment to investigate the effect of varying plasmid concentration on the efficiency of *E. coli* transformation.

You may, or otherwise, also use the following equipment and materials:

- Solution of plasmid containing ampicillin resistance gene of concentration 100 Unit/ml
- Suspension of competent *E. coli* cells
- Distilled water
- LB broth (nutrient solution)
- Petri dishes containing LB agar medium and ampicillin (LB/amp plates)
- Sterilised microfuge tubes
- Micropipettes
- Bleach solution
- alcohol

Your plan should have a clear and helpful structure to include:

- an explanation of theory to support your practical procedure
- a description of the method used including scientific reasoning behind the method
- an explanation of the dependent and independent variables involved
- relevant, clearly labelled diagrams
- how you will record your results and ensure they are as accurate and reliable as possible
- proposed layout of results tables and graphs with clear headings and labels
- correct use of scientific and technical terms

[Total:12]

Free-response question

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 (a) Rubisco is an important enzyme in the Calvin cycle. [6]

Outline how it might be possible to produce genetically modified plants with increased rubisco concentrations, or rubisco with enhanced efficiency.

(b) Explain the basis of genetic engineering and how genetic engineering can improve the yield of one named crop plant and one named animal in solving the demand for food in the world. [8]

(c) Discuss how the human genome project can help parents customise their babies and the possible implications of doing so. [6]

[Total: 20]